

The Rose Operator: Bridging Composition Maps and Transfer Kernels via Local Linear Models

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1 Local Linearizations

Consider the Cauchy problem

$$\begin{aligned}\frac{du}{dt} + A(u(t)) &= 0, \quad t > 0, \\ u(0) &= u_0, \quad u_0 \in U \subset X,\end{aligned}\tag{1}$$

where A is an operator (possibly nonlinear and state-dependent) that generates a contraction semi-group $T(h)$ on the Banach space $\mathcal{X}$.

In many practical settings, one only has access to discrete observations of the state. Let these observations be arranged in the data matrix

$$X_t = \begin{bmatrix} x(t) \\ x(t-h) \\ x(t-2h) \\ \vdots \end{bmatrix},\tag{2}$$

where the rows correspond to states sampled at time intervals of length h .

A first approach is to attempt a global linearization via a least-squares formulation:

$$X_t \approx X_{t-h} T(h),\tag{3}$$

$$\text{or equivalently, } u(t) \approx T(h) x(t-h).\tag{4}$$

If the residuals from this approximation are well within an acceptable and prescribed tolerance—then it is reasonable to assume that the dynamics are well represented by a linear operator. In such a case, one can recover the global generator via

$$A = \lim_{h \rightarrow 0^+} \frac{1}{h} (T(h) - I).\tag{5}$$

This approach is particularly effective when A is linear or nearly linear. The contraction property aids in establishing the existence and uniqueness of solutions, however it also means that a global linearization of A may be unattainable when its dependence on $u(t)$ is significant with residuals remaining unacceptably large.

A global linearization may fail to capture the system's behavior when A is state dependent. In such case a local linearization may be preferable.

To construct a local approximation we can introduce a weighting operator W designed to select or emphasize those rows in X_t that are dynamically relevant. The idea is to filter the data so that regions exhibiting dynamics similar to those surrounding $x(t)$ are highlighted, typically by considering observations local to $x(t)$ under some metrization of $\mathcal{X}$.

If W is constructed using a localization kernel, $k_\theta(x(j), x(i))$, where θ parametrizes what “local” means, then the weighted observations are,

$$W_\theta X_t,\tag{6}$$

and the local approximation is expressed as

$$W_\theta X_t \approx W_\theta X_{t-h} \hat{T}(h, \theta, t), \quad (7)$$

where $\hat{T}(h, \theta, t)$ is the locally linearized evolution operator valid in a neighborhood of $x(t)$. If a weighted pseudoinverse is employed, one may write

$$A(x(t), \theta, h) = \left((W_\theta X_{t-h})^+ (W_\theta X_t) - I \right) \quad (8)$$

where $(W X_{t-h})^+$ denotes the pseudoinverse of $W X_{t-h}$. Now recovery of the generator $A(x(t))$ requires an additional spatial limit.

$$A(x(t)) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\lim_{\theta \rightarrow 0} \hat{T}(h, \theta, t) - I \right) \quad (9)$$

If T is (i) deterministic (i.e. $\theta \rightarrow 0$ iff $k_\theta \rightarrow \delta$), and (ii) locally linear (i.e. $T = \sum_{k=0}^{\infty} \frac{t^k}{k!} A(x_j, \theta, h)^k$ converges where required), then this is a nice framework for approximating the state-dependent generator $A(x(t))$ in the presence of features precluding a global linear model.

However, in the absence of any strong *a priori* intuition regarding even the most general assumptions of the generating process, most notably stochasticity and the rate of decay of higher order terms, deciding how to approach these limits (and thus what constitutes a *good* linearization) is not easy.

2 Locality in Infinitesimal Generators

2.1 The Evolution Kernel

In the classical theory of C_0 semigroups, the evolution operator $T(h)$ composes a C_0 semigroup generated by a linear operator A

$$T(h) = e^{hA}.$$

In this setting, the kernel $K_h(\xi, \eta)$ is the Dirac delta

$$K_h(\xi, \eta) \xrightarrow{h \rightarrow 0} \delta(\xi - \eta).$$

Proof. Consider the integral operator

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta \quad (10)$$

with infinitesimal generator

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} (T(h) - I) f(\xi) \quad (11)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} ((T(h)f)(\xi) - f(\xi)) \quad (12)$$

By substitution

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta - f(\xi) \right) \quad (13)$$

The goal is to rewrite the integral term so that $f(\xi)$ appears inside the integral. We add and subtract $K_h(\xi, \eta)f(\xi)$ inside the integral,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) - K_h(\xi, \eta) f(\xi) + K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (14)$$

and factor,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \int_{\Omega} K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (15)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) \quad (16)$$

We can ensure that $K_h(\xi, \eta)$ acts as a state transition probability density by imposing the normalization condition,

$$\int_{\Omega} K_h(\xi, \eta) d\eta = 1 \quad (17)$$

while at the same time eliminating the second term,

$$\lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) = 0 \quad (18)$$

thus recovering the infinitesimal generator as a standard non-local operator,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta \quad (19)$$

Kernel Structure: The resolvent operator associated with A is given by:

$$R_{\lambda}(A) = (\lambda I - A)^{-1}. \quad (20)$$

It is well known (see, e.g. Pazy) that the semigroup $T(h)$ can be expressed in terms of the resolvent via the contour integral,

$$T(h) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_{\lambda}(A) d\lambda \quad (21)$$

We equate the integral representations

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_{\lambda}(A) f(\xi) d\lambda \quad (22)$$

Furthermore, Arendt & Bukhvalov showed that if $T(h)$ is an integral operator, then so is the resolvent,

$$(R_\lambda f)(\xi) = \int_{\Omega} M_\lambda(\xi, \eta) f(\eta) d\eta \quad (23)$$

where $M_\lambda(\xi, \eta)$ is the resolvent kernel. From (22) using the integral resolvent,

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \int_{\Omega} M_\lambda(\xi, \eta) f(\eta) d\eta d\lambda \quad (24)$$

Two cases follow regarding the absolute integrability of

$$F(\lambda, \eta) = e^{h\lambda} M_\lambda(\xi, \eta) f(\eta)$$

If $F \in L^1(\Gamma \times \Omega)$ then for

$$\oint_{\Gamma} \int_{\Omega} |F(\lambda, \eta)| d\mu(\eta) d\nu(\lambda) \quad (25)$$

we may apply Fubini's theorem and interchange the order of integration

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \int_{\Omega} \left(\frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_\lambda(\xi, \eta) d\lambda \right) f(\eta) d\eta \quad (26)$$

Since this must hold for all f , we identify the kernels,

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_\lambda(\xi, \eta) d\lambda \quad (27)$$

Since $T(0) = I$, the kernel at $h = 0$ must satisfy

$$K_0(\xi, \eta) = \delta(\xi - \eta) \quad (28)$$

In many classical settings, one can show that the resolvent satisfies

$$(\lambda I - A)R_\lambda(A) = I \quad (29)$$

so that in kernel form

$$(\lambda I - A)M_\lambda(\xi, \eta) = \delta(\xi - \eta) \quad (30)$$

$M_\lambda(\xi, \eta)$ serves as the Green's function for the operator $\lambda I - A$. Supposing that for values of λ in a suitable region the Neumann series expansion hold

$$(\lambda I - A)^{-1} = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \quad (31)$$

Then the kernel $M_\lambda(\xi, \eta)$ may be expressed as

$$M_\lambda(\xi, \eta) = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \quad (32)$$

$$M_\lambda(\xi, \eta) = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \quad (33)$$

Substitute this expansion into

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \left(\sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \right) d\nu(\lambda) \quad (34)$$

$$= \delta(\xi - \eta) \sum_{n=0}^{\infty} A^n \left[\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) \right] \quad (35)$$

According to the residue theorem

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) = \frac{h^n}{n!} \quad (36)$$

Substituting this back into (35), we obtain

$$K_h(\xi, \eta) = \delta(\xi - \eta) \sum_{n=0}^{\infty} \frac{h^n}{n!} A^n = \delta(\xi - \eta) e^{hA} \quad (37)$$

Thus,

$$K_h(\xi, \eta) = \delta(\xi - \eta) e^{hA} \quad (38)$$

which is equivalent to the operator identity,

$$T(h) = e^{hA} \quad (39)$$

□

The Nonlinear (State-Dependent) Case: When the system is nonlinear, the evolution is typically given by a state-dependent mapping,

$$T(h)(\xi) = \Phi_h(\xi),$$

we may still maintain an integral representation:

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta.$$

In order for the kernel to capture the correct local behavior of the flow, we require that for small h ,

$$K_h(\xi, \eta) \rightarrow \delta(\Phi_h(\xi) - \Phi_h(\eta)).$$

Since Φ_0 is the identity mapping, this condition reduces to

$$\lim_{h \rightarrow 0} K_h(\xi, \eta) = \delta(\xi - \eta),$$

which, in the weak sense, means that for any test function f ,

$$\lim_{h \rightarrow 0} \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = f(\xi).$$

3 Locally-linear Stochastic Operators

3.1 Local linear model

For each embedded state $\mathbf{x} \in \mathcal{X}$ fit a local globally weighted least-squares linear map

$$A_\theta(\mathbf{x}) = \arg \min_A \sum_{j \in \mathcal{N}_\theta(\mathbf{x})} w_\theta(\mathbf{x}, \mathbf{x}_j) \|\mathbf{x}_{j+1} - A \mathbf{x}_j\|^2 \quad (40)$$

where $\mathcal{N}_\theta(\mathbf{x})$ is the neighbourhood chosen with bandwidth (or kernel scale) θ and

$$w_\theta(\mathbf{x}, \mathbf{x}_j) = \exp\left(-\theta \|\mathbf{x} - \mathbf{x}_j\| / \|\mathbf{x}_1 - \mathbf{x}_j\|\right) \quad (41)$$

The residuals

$$\mathbf{r}_j = \mathbf{x}_{j+1} - A_\theta(\mathbf{x}) \mathbf{x}_j \quad j \in \mathcal{N}_\theta(\mathbf{x}) \quad (42)$$

yield an empirical covariance

$$\Sigma_\theta(\mathbf{x}) = \text{Cov}_{j \in \mathcal{N}_\theta(\mathbf{x})}[\mathbf{r}_j] \quad (43)$$

3.2 Stochastic transition kernel

Replacing the Dirac mass of a deterministic map by a Gaussian centred on the local linear prediction gives

$$k_\theta(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(\mathbf{y}; A_\theta(\mathbf{x}) \mathbf{x}, \Sigma_\theta(\mathbf{x})) \quad (44)$$

a data-driven transition density that captures both drift and anisotropic noise. A data driven stochastic Perron-Frobenius operator is given by the action of the kernel on probability densities ρ ,

$$(P_\theta \rho)(\mathbf{y}) = \int_{\mathcal{M}} k_\theta(\mathbf{y} \mid \mathbf{x}) \rho(\mathbf{x}) d\mathbf{x} \quad (45)$$

3.3 Continuous-time interpretation

For small sampling interval Δt the update

$$\mathbf{x}_{t+\Delta t} = A_\theta(\mathbf{x}_t) \mathbf{x}_t + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_\theta(\mathbf{x}_t)) \quad (46)$$

is a locally linear Itô SDE; subtract $\mathbf{x}_t$ and divide by the step size,

$$\frac{\mathbf{x}_{t+\Delta t} - \mathbf{x}_t}{\Delta t} = \left(\frac{A_\theta(\mathbf{x}_t) - I}{\Delta t} \right) \mathbf{x}_t + \frac{\boldsymbol{\eta}_t}{\Delta t} \quad (47)$$

Write the noise in the *diffusion scaling*

$$(48)$$

$$\boldsymbol{\eta}_t = C(\mathbf{x}_t)\sqrt{\Delta t} \boldsymbol{\xi}_t \quad \boldsymbol{\xi}_t \sim \mathcal{N}(0, I) \quad (49)$$

so that

$$\Sigma_\theta(\mathbf{x}_t) = C(\mathbf{x}_t)C^\top(\mathbf{x}_t)\Delta t \quad (50)$$

Define the local drift matrix

$$B(\mathbf{x}) = A_\theta(\mathbf{x}) - I \quad (51)$$

Taking $\Delta t \rightarrow 0$ yields the Itô SDE

$$d\mathbf{X}_t = B(\mathbf{X}_t) \mathbf{X}_t dt + C(\mathbf{X}_t) d\mathbf{W}_t \quad (52)$$

where $\mathbf{W}_t$ is a standard Wiener process. The equation is *state-dependent linear*: for fixed x the map $B(x)x$ is linear, but the coefficients vary smoothly over the manifold.

If one keeps Σ_θ *independent* of Δt , then the result is a discrete-time Markov chain rather than the time discretization of the continuous case. Both views preserve the central structure:

$$\text{“locally linear drift”} + \text{“additive Gaussian innovation”}$$

Defining $B(\mathbf{x}) = A_\theta(\mathbf{x}) - I$ and $C(\mathbf{x})C^\top(\mathbf{x}) = \Sigma_\theta(\mathbf{x})/\Delta t$, the infinitesimal generator acting on a smooth test function f is

$$(\mathcal{L}f)(\mathbf{x}) = \nabla f(\mathbf{x})^\top B(\mathbf{x}) \mathbf{x} + \frac{1}{2} \text{tr}(\Sigma_\theta(\mathbf{x}) \nabla^2 f(\mathbf{x})). \quad (53)$$

3.4 Connection to Wold’s Decomposition

For a covariance-stationary scalar process X_t , Wold’s theorem states

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} + \eta_t,$$

where $\{\varepsilon_t\}$ are orthogonal *innovations*, $\psi_0 = 1$, $\sum \psi_j^2 < \infty$, and η_t is a perfectly predictable (finite-rank) component.

In the locally-linear framework we obtain an analogous, *state-dependent* split at every embedded point $\mathbf{x}_t$:

$$\underbrace{\mathbf{x}_{t+1}}_{\text{next state}} = \underbrace{A_\theta(\mathbf{x}_t) \mathbf{x}_t}_{\text{predictable drift}} + \underbrace{\varepsilon_t(\theta)}_{\text{innovation}}, \quad \varepsilon_t(\theta) = \mathbf{x}_{t+1} - A_\theta(\mathbf{x}_t) \mathbf{x}_t.$$

3.5 Fitting Algorithm

Consider observations of state transition pairs $(y_i, x_i)_{i \in N}$ where $y_i = x_{i+1}$.

Algorithm 3.1 (H). Data-Driven Delay Embedding and Pointwise θ Selection via Rose+PCA

Require: Univariate series $\{x(t)\}$, over-embed dimension D , grid size K , variance threshold $\alpha \in (0, 1)$.

Ensure: Reduced dimension d , per-anchor $\{\theta_i^*\}$, final residuals $\{\tilde{r}_i \in \mathbb{R}^d\}$, coefficients $\{\tilde{C}_i \in \mathbb{R}^{d \times d}\}$.

```

1: 1. Build full  $D$ -dimensional delay embedding:
2: for  $i = 1$  to  $N$  do
3:    $\mathbf{x}_i \leftarrow [x(t_i), x(t_i - 1), x(t_i - 2), \dots, x(t_i - (D - 1))]$   $\in \mathbb{R}^D$ .
4: end for
5:  $E \leftarrow [\mathbf{x}_1; \dots; \mathbf{x}_N] \in \mathbb{R}^{N \times D}$ .
6: 2. Construct  $\theta$ -grid
7:  $\theta \leftarrow [\theta_0 = 0; \dots; \theta_K = \theta_{max}]$ 
8: 4. Compute Rose residuals  $r_{i,k} \in \mathbb{R}^D$ :
9: for  $i = 1$  to  $N$  do
10:   for  $k = 1$  to  $K$  do
11:     Fit WLS at  $\mathbf{x}_i$  with kernel scale  $\theta_k$ , forecast next step  $\hat{\mathbf{x}}_i$ .
12:      $r_{i,k} \leftarrow \mathbf{x}(t_i + 1) - \hat{\mathbf{x}}_i$ .
13:   end for
14: end for
15: 5. Select pointwise  $\theta_i^*$ :
16: for  $i = 1$  to  $N$  do
17:    $\rho_i(k) \leftarrow \|r_{i,k}\|_2, \quad k_i^* \leftarrow \arg \min_{1 \leq k \leq K} \rho_i(k)$ .
18:    $\theta_i^* \leftarrow \theta_{k_i^*}$ .
19: end for
20: 6. Collect selected residuals in  $\mathbb{R}^D$ :
21: for  $i = 1$  to  $N$  do
22:    $r_i^* \leftarrow r_{i, k_i^*}$ .
23: end for
24:  $R^* \leftarrow [r_1^*; \dots; r_N^*] \in \mathbb{R}^{N \times D}$ .
25: 7. PCA on  $R^*$  to determine reduced dimension  $d$ :
26: Compute SVD:  $R^* = U \Sigma V^\top$ .
27: Choose smallest  $d$  with  $\sum_{j=1}^d \sigma_j^2 / \sum_{j=1}^D \sigma_j^2 \geq \alpha$ .
28:  $V_d \leftarrow V[:, 1 : d] \in \mathbb{R}^{D \times d}$ .
29: 8. Form reduced embedding in  $\mathbb{R}^d$ :
30: for  $i = 1$  to  $N$  do
31:    $\tilde{\mathbf{x}}_i \leftarrow V_d^\top \mathbf{x}_i \in \mathbb{R}^d$ .
32: end for
33: 9. Re-fit Rose in  $\mathbb{R}^d$  at  $\theta_i^*$ :
34: for  $i = 1$  to  $N$  do
35:   Fit WLS on  $\tilde{\mathbf{x}}_i$  with scale  $\theta_i^*$ , forecast  $\hat{\tilde{\mathbf{x}}}_i$ .
36:    $\tilde{r}_i \leftarrow \tilde{x}(t_i + 1) - \hat{\tilde{\mathbf{x}}}_i \in \mathbb{R}^d$ .
37:   Extract coefficient  $\tilde{C}_i \in \mathbb{R}^{d \times d}$ .
38: end for
39: 10. Save outputs via save_all:
40:  $\{\tilde{r}_i\}_{i=1}^N, \{\tilde{C}_i\}_{i=1}^N, \{t_i\}_{i=1}^N, \{\theta_i^*\}_{i=1}^N$ .

```

4 Multiscale Diagnostics with the Rose Operator

The Rose operator gives a one-parameter family of transfer operators that varies continuously from the deterministic Koopman pull-back ($\theta \rightarrow 0$) to a maximally diffusive Perron–Frobenius push-forward ($\theta \rightarrow \infty$). Here $C_x^{(\theta)}$ is the locally-weighted least-squares estimate of the Jacobian at x , while $p_x^{(\theta)}$ is a kernel-density estimate of the residuals. Adjusting the bandwidth θ thus performs a *multiscale experiment* that helps disentangle strong non-linearity from irreducible noise.

4.1 $\mathcal{R}_\theta$ as a continuum bridge

The Rose operator generates a one-parameter *continuous path of transfer operators*

$$\mathcal{R}_\theta : f \mapsto \int_{\mathbb{R}^d} f(x') k_\theta(x' | x) dx', \quad k_\theta(x' | x) = p_x^{(\theta)}(x' - C_x^{(\theta)} x),$$

indexed by a bandwidth $\theta > 0$. In what follows we write $w_\theta(r) = \exp(-r^2/\theta^2)$ for the Gaussian weight,

$$C_x^{(\theta)} = \arg \min_{C \in \mathbb{R}^{d \times d}} \sum_{s=0}^{N-1} w_\theta(\|x_s - x\|) \|x_{s+1} - C x_s\|^2,$$

and let $p_x^{(\theta)}$ be the KDE of the residuals $r_s^{(\theta)} = x_{s+1} - C_x^{(\theta)} x_s$ collected inside the θ -ball centred at x .

Assumption 4.1 (Flow regularity). The data $\{x_t\}_{t=0}^N$ are samples of a C^2 diffeomorphism $\varphi^{\delta t} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ observed with i.i.d. additive noise of mean 0 and covariance $\sigma^2 I_d$ (the deterministic case is covered by $\sigma = 0$).

Theorem 4.1 (Rose $\Rightarrow$ Koopman–PF bridge). *Under Assumption 4.1 the following hold.*

(i) Deterministic limit. For every $f \in C_b^1(\mathbb{R}^d)$,

$$\|\mathcal{R}_\theta f - (f \circ D\varphi^{\delta t})\|_{L^\infty} = \mathcal{O}(\theta^2) + \mathcal{O}((N\theta^d)^{-1/2}) \quad \text{as } \theta \downarrow 0 \quad (54)$$

i.e. $\mathcal{R}_\theta \rightarrow K$ in operator norm when $\theta \rightarrow 0$ and $N\theta^d \rightarrow \infty$.

(ii) Diffusive limit. Let $p_\infty(\xi) = \lim_{\theta \rightarrow \infty} p_x^{(\theta)}(\xi)$ (the empirical residual density averaged over x). Then

$$\lim_{\theta \rightarrow \infty} \mathcal{R}_\theta f = \int_{\mathbb{R}^d} f(x') p_\infty(x') dx' = \mathbb{E}_{p_\infty}[f] \quad (55)$$

a rank-one Perron–Frobenius operator corresponding to maximal diffusion.

(iii) Continuity in θ . For any $0 < \theta_1 < \theta_2$ one has

$$\|\mathcal{R}_{\theta_1} - \mathcal{R}_{\theta_2}\|_{L^2 \rightarrow L^2} \leq C |\theta_1 - \theta_2|, \quad (56)$$

where C depends only on $\sup_x \|D^2 \varphi^{\delta t}(x)\|$ and on tail moments of the residuals.

Proof sketch. (i) Local-linear regression of a C^2 map with isotropic Gaussian weights has bias $\mathcal{O}(\theta^2)$ and variance $\mathcal{O}((N\theta^d)^{-1})$ [?, Chap. 6]. Plugging these into k_θ yields the stated norm bound.

(ii) For $\theta \rightarrow \infty$ the weights become uniform, so $C_x^{(\theta)} \rightarrow C_\infty$ (global least-squares) and $p_x^{(\theta)} \rightarrow p_\infty$, losing all dependence on x .

(iii) Both $C_x^{(\theta)}$ and the bandwidth of $p_x^{(\theta)}$ are Lipschitz in θ under Assumption 4.1; integrate the pointwise bound over x . $\square$

Remark 4.1. Statement (i) collapses to the *exact* Koopman operator when the flow is already linear and the measurement noise satisfies $\sigma = 0$. Statement (ii) represents the opposite extreme of maximal diffusion. Together with (iii) they show that $\theta \mapsto \mathcal{R}_\theta$ forms a **continuum bridge** between these dynamical operators, making θ an explicit scale parameter for separating strong non-linearity (bias $\propto \theta^2$) from irreducible noise (variance floor σ^2).

4.2 Residual-scaling law

Define the mean-square residual

$$\varepsilon(\theta)^2 = \frac{1}{N} \sum_{t=0}^{N-1} \|x_{t+1} - C_{x_t}^{(\theta)} x_t\|^2. \quad (57)$$

For a C^2 deterministic flow one obtains the bias-variance decomposition

$$\varepsilon(\theta)^2 = A\theta^4 + \frac{B}{N\theta^d} + \sigma^2, \quad (58)$$

where $A, B > 0$ depend on higher-order derivatives and the sample distribution. The constant term σ^2 vanishes for purely deterministic dynamics and equals the noise variance when a stochastic component is present. Plotting $\varepsilon(\theta)$ in log-log coordinates therefore yields a clear diagnostic: absence (presence) of a horizontal floor at small θ indicates deterministic (stochastic) residuals.

4.3 Covariance-floor test

Let

$$\Sigma^{(\theta)}(x) = \mathbb{E}[r^{(\theta)} r^{(\theta)\top} \mid x] \quad (59)$$

be the conditional covariance of the residuals. For deterministic errors $\lambda_{\max}(\Sigma^{(\theta)}(x)) = \mathcal{O}(\theta^4)$, whereas additive or multiplicative noise produces a positive-definite floor independent of θ . Tracking the largest eigenvalue across a ladder of bandwidths and anchor points therefore produces a spatial noise map.

4.4 Algorithmic workflow

Algorithm 4.1 summarizes the practical steps required to apply these diagnostics. Section ?? demonstrates their effectiveness on synthetic data with known ground truth.

Algorithm 4.1 (t). Rose-operator diagnostics

Require: data pairs $\{(x_t, x_{t+1})\}_{t=0}^{N-1}$; bandwidth ladder $\Theta = \{\theta_k\}_{k=0}^K$

- 1: **for** θ **in** Θ **do**
- 2: Compute $C_{x_t}^{(\theta)}$ and residuals $r_t^{(\theta)} = x_{t+1} - C_{x_t}^{(\theta)} x_t$
- 3: Update $\varepsilon(\theta)^2$ and the empirical covariances $\Sigma^{(\theta)}(x_t)$
- 4: **end for**
- 5: Plot $\theta \mapsto \varepsilon(\theta)$ and $\theta \mapsto \lambda_{\max}(\Sigma^{(\theta)}(x))$
- 6: Infer the presence or absence of irreducible noise from the small- θ behaviour (cf. (58))

5 A Categorical Scaffold for Local Linearization

Local linear models, residual covariances, and adaptive basis updates can all be written algebraically, but a categorical formulation clarifies *which parts belong to the dynamics, which to the coordinate representation, and which to the data-driven closure*. By lifting the construction to functors, natural transformations, and (co)monads we obtain

- (i) a precise separation of concerns
- (ii) a route to operator-theoretic results via functoriality
- (iii) and a universal property (*cofree* coalgebra) that characterises the residual-frame augmentation

5.1 Components

Throughout, $k \geq 2$ is fixed, so that C^k flows have well-defined first and second jets. All manifolds are assumed compact, and all vector spaces are over $\mathbb{R}$.

Definition 5.1 (Dyn). The category **Dyn** consists of

- **objects** $(\mathcal{M}, \varphi^t)$ where $\mathcal{M}$ is a compact C^k manifold and $\varphi^t : \mathcal{M} \rightarrow \mathcal{M}$ a C^k flow.
- **morphisms** $\Psi : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t)$ given by C^k maps $\Psi : \mathcal{M} \rightarrow \mathcal{N}$ satisfying

$$\Psi \circ \varphi^t = \psi^t \circ \Psi \quad (60)$$

for all $t \in \mathbb{R}$.

Remark 5.1. Objects in **Dyn** are smooth dynamical systems. Morphisms Ψ are *dynamic-preserving* (equivalently, *equivariant* or *trajectory-intertwining*). They are dynamic-preserving in the sense that for smooth manifolds and flows:

$$(\mathcal{M}, \varphi^t), (\mathcal{N}, \psi^t) \quad t \in \mathbb{R} \quad (61)$$

the morphism,

$$\Psi = (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t) \quad (62)$$

is a C^k ($k \geq 1$) map that is equivariant,

$$\Psi(\varphi^t(x)) = \psi^t(\Psi(x)) \quad \forall x \in \mathcal{M} \quad (63)$$

so that Ψ transports trajectories φ^t onto trajectories ψ^t :

$$\Psi \circ \varphi^t = \psi^t \circ \Psi \quad (64)$$

Example 5.1. Take the two-torus with linear flow,

$$(\mathbb{T}^2, \varphi^t), \quad \varphi^t(x, y) = (x + \alpha t, y + \beta t) \pmod{1} \quad (65)$$

and the circle with rigid rotation,

$$(\mathbb{S}^1, \psi^t), \quad \psi^t(\theta) = \theta + \alpha t \pmod{1} \quad (66)$$

Let,

$$\Psi : \mathbb{T}^2 \rightarrow \mathbb{S}^1, \quad \Psi(x, y) = x \pmod{1} \quad (67)$$

For all $(x, y) \in \mathbb{T}^2$ $t \in \mathbb{R}$:

$$\Psi(\varphi^t(x, y)) = \Psi(x + \alpha t, y + \beta t) \quad (68)$$

$$= (x + \alpha t) \pmod{1} \quad (69)$$

$$= \psi^t(\Psi(x, y)) \quad (70)$$

so,

$$\Psi \circ \varphi^t = \psi^t \circ \Psi, \quad \forall t \in \mathbb{R} \quad (71)$$

and Ψ is a dynamic-preserving C^∞ map and morphism of **Dyn**.

Remark 5.2 (Discrete-time systems via suspension). Although **Dyn** is defined only for *flows*, any discrete-time diffeomorphism $T : \mathcal{M} \rightarrow \mathcal{M}$ can be embedded in **Dyn** through its *suspension flow*.

Form the mapping torus

$$\text{Sus}(\mathcal{M}, T) := (\mathcal{M} \times [0, 1]) / (x, 1) \sim (Tx, 0), \quad (72)$$

and define the flow

$$\varphi^t[(x, u)] = [(T^{\lfloor t+u \rfloor} x, u + t - \lfloor t+u \rfloor)] \quad (73)$$

Then $(\text{Sus}(\mathcal{M}, T), \varphi^t)$ is an object of **Dyn** whose *time-1 map* reproduces T :

$$\varphi^1[(x, 0)] = [(Tx, 0)] \quad (74)$$

This construction extends to morphisms by $\Sigma(f)[(x, u)] = [(f(x), u)]$, producing a functor $\Sigma : \mathbf{Dyn}_{\mathbb{Z}} \rightarrow \mathbf{Dyn}$. As such, results for continuous-time objects apply *mutatis mutandis* to discrete systems via Σ .

Definition 5.2 (Vect). The category **Vect** consists of

- **objects** V finite-dimensional real vector spaces.
- **morphisms** $L : V \rightarrow W$ linear maps.

Definition 5.3 (Embedding functor). Fix, for every Dyn-object $(\mathcal{M}, \varphi^t)$ a generic embedding $\Phi_{\mathcal{M}} : \mathcal{M} \hookrightarrow \mathbb{R}^{m(\mathcal{M})}$. Define the functor

$$F : \mathbf{Dyn} \longrightarrow \mathbf{Vect} \quad \text{by} \quad \begin{cases} F(\mathcal{M}, \varphi^t) = \mathbb{R}^{m(\mathcal{M})}, \\ F(\Psi) = \Phi_{\mathcal{N}} \circ \Psi \circ \Phi_{\mathcal{M}}^{-1} : \mathbb{R}^{m(\mathcal{M})} \longrightarrow \mathbb{R}^{m(\mathcal{N})}. \end{cases}$$

with identity $F(\text{id}_{(\mathcal{M}, \varphi^t)}) = \text{id}_{\mathbb{R}^{m(\mathcal{M})}}$.

Functoriality is immediate from $\Phi_{\mathcal{N}} \circ (\Psi_2 \circ \Psi_1) = F(\Psi_2) \circ F(\Psi_1)$, which gives the following diagram:

$$\begin{array}{ccc} (\mathcal{M}, \varphi^t) & \xrightarrow{\Phi := \Phi_{\mathcal{M}}} & \mathbb{R}^{m(\mathcal{M})} \\ \Psi \downarrow & & \downarrow \Phi' \circ \Psi \circ \Phi^{-1} \\ (\mathcal{N}, \psi^t) & \xrightarrow{\Phi' := \Phi_{\mathcal{N}}} & \mathbb{R}^{m(\mathcal{N})} \end{array}$$

Definition 5.4 (Dyn Component-wise product). Let $(\mathcal{M}_i, \varphi_i^t)_{i=1}^n$ be objects of **Dyn**. Define

$$\mathcal{M} := \prod_{i=1}^n \mathcal{M}_i, \quad \varphi^t(x_1, \dots, x_n) := (\varphi_1^t(x_1), \dots, \varphi_n^t(x_n)), \quad (75)$$

and projection maps $\pi_j(x_1, \dots, x_n) = x_j$ ($j = 1, \dots, n$). We call

$$(\mathcal{M}, \varphi^t) \quad (76)$$

the *component-wise product* of the given objects.

Lemma 5.1 (Universal property of finite products in **Dyn**). *For the system $(\mathcal{M}, \varphi^t)$ and projections π_j defined above, the pair*

$$((\mathcal{M}, \varphi^t), \{\pi_j\}_{j=1}^n) \quad (77)$$

*is the categorical product of $(\mathcal{M}_i, \varphi_i^t)_{i=1}^n$ in the category **Dyn**.*

Proof. Because each π_j is smooth,

$$\pi_j(\varphi^t(x_1, \dots, x_n)) = \varphi_j^t(x_j) = \varphi_j^t(\pi_j(x_1, \dots, x_n)), \quad (78)$$

so π_j intertwines the flows and is therefore a morphism in **Dyn**.

Let $(\mathcal{N}, \psi^t)$ be any object and suppose we are given morphisms $f_j : (\mathcal{N}, \psi^t) \rightarrow (\mathcal{M}_j, \varphi_j^t)$ for $j = 1, \dots, n$. Define

$$f : \mathcal{N} \longrightarrow \mathcal{M}, \quad f(y) := (f_1(y), \dots, f_n(y)). \quad (79)$$

The map f is C^k because each f_j is. Equivariance follows component-wise:

$$f(\psi^t y) = (f_1(\psi^t y), \dots) = (\varphi_1^t f_1(y), \dots) = \varphi^t(f(y)). \quad (80)$$

Hence f is a morphism in **Dyn** and $\pi_j \circ f = f_j$ by construction.

If $g : (\mathcal{N}, \psi^t) \rightarrow (\mathcal{M}, \varphi^t)$ is another morphism with the same property $\pi_j \circ g = f_j$ for every j , then for each $y \in \mathcal{N}$

$$g(y) = (\pi_1 g(y), \dots, \pi_n g(y)) = (f_1(y), \dots, f_n(y)) = f(y),$$

so $g = f$.

Since a unique mediating morphism exists for every candidate cone, $(\mathcal{M}, \varphi^t)$ with the projections π_j satisfies the universal property of finite products in **Dyn**. $\square$

Lemma 5.2 (Coproducts). *For any family $(\mathcal{M}_i, \varphi_i^t)_{i \in I}$, the disjoint union*

$$\left(\bigsqcup_{i \in I} \mathcal{M}_i, \varphi^t(x) = \varphi_i^t(x) \text{ if } x \in \mathcal{M}_i \right)$$

*with the canonical injections $\iota_i : \mathcal{M}_i \hookrightarrow \bigsqcup_j \mathcal{M}_j$ is the coproduct in **Dyn**.*

Proof. Disjoint unions of compact manifolds are compact manifolds, the flow is C^k on each component, and the injections are equivariant by construction. The universal property follows from the universal property of disjoint unions in smooth manifolds. $\square$

Lemma 5.3 (Regular equalisers). *Let $f, g : (\mathcal{M}, \varphi^t) \rightarrow (\mathcal{N}, \psi^t)$ be morphisms. Assume f and g are transverse to the diagonal $\Delta_{\mathcal{N}} \subset \mathcal{N} \times \mathcal{N}$. Then*

$$E := \{x \in \mathcal{M} \mid f(x) = g(x)\}$$

*is an embedded C^k sub-manifold, invariant under φ^t . The inclusion $i : E \hookrightarrow \mathcal{M}$ with the restricted flow $\varphi|_E^t$ realises the equaliser of (f, g) in **Dyn**.*

Proof. Transversality implies E is an embedded sub-manifold. Equivariance of f and g gives $f(\varphi^t x) = \psi^t f(x) = \psi^t g(x) = g(\varphi^t x)$, so E is flow-invariant. The inclusion i is a morphism, and any morphism $h : (\mathcal{K}, \chi^t) \rightarrow (\mathcal{M}, \varphi^t)$ with $fh = gh$ factors uniquely through i by the usual smooth-manifold argument. $\square$

Remark 5.3 (Limitations of completeness). The category of C^k manifolds is neither complete nor cocomplete; an infinite product need not be a manifold, and general coequalisers (or non-regular equalisers) can fail to carry a smooth Hausdorff structure. Consequently **Dyn** is *finitely* complete and has coproducts, but it does not possess arbitrary small limits or colimits.

Lemma 5.4 (Limits and colimits). ***Dyn** has all small limits and colimits; they are inherited point-wise from the category of C^k manifolds.*

Sketch. Products exist because finite products of compact manifolds are compact manifolds; the flow acts component-wise. Equalisers are submanifolds defined by equalising maps that commute with the flow. Dually for colimits. $\square$

The remainder of this section builds functors *out of* **Dyn** and uses natural transformations and comonads to formalise local linearisation and its residual correction.

5.2 Local linearisation as a natural transformation

Write $\Pi_d : \mathbf{Dyn} \rightarrow \mathbf{Dyn}$ for the functor that forgets all ≥ 2 -jet data. Weighted least squares at each anchor x induces the component

$$\eta_{(\mathcal{M}, \varphi^t)}(x) := B_x : F(\mathcal{M}, \varphi^t) \longrightarrow F \circ \Pi_d(\mathcal{M}, \varphi^t),$$

yielding a natural transformation $\eta : F \Rightarrow F \circ \Pi_d$.

5.3 Residual frame field as a cofree coalgebra

Define a comonad $\mathbf{G} : (\mathbf{Vect}, \otimes) \rightarrow (\mathbf{Vect}, \otimes)$ by $\mathbf{G}(V) = V \oplus (\text{span } Q_x)$. The map $x \mapsto (B_x, Q_x)$ endows each fibre with the structure of a $\mathbf{G}$ -coalgebra, cofree under the variance-fraction constraint ρ . In string-diagram notation:

$$\begin{array}{ccc} F(\mathcal{M}, \varphi^t) & \xrightarrow{\quad \Delta \quad} & \mathbf{G}F(\mathcal{M}, \varphi^t) \\ \eta \downarrow & & \downarrow \mathbf{G}\eta \\ F \circ \Pi_d(\mathcal{M}, \varphi^t) & \xrightarrow{\quad \mathbf{G}\Delta \quad} & \mathbf{G}F \circ \Pi_d(\mathcal{M}, \varphi^t) \end{array}$$

The dashed arrows are the Rose augmentation; commutativity is enforced by letting $\mathbf{G}$ be *cofree* with respect to ρ .

References

A Notation & Conventions

PX	probability measures on $(X, \mathcal{F})$
G	Giry monad (B.15)
$x \mapsto \delta_x$	Dirac embedding
$\rightsquigarrow$	kernel arrow (Def. B.13)
$\flat, \sharp$	musical conversions (Def. ??)
P_t, L	Markov semigroup & generator (B.23–B.24)
k_Δ, T	LWLS kernel & Rose operator (??–??)
Δ	discrete time step in data

B Definitions

B.1 Measure & Probability Foundations

Definition B.1 (σ -Algebra). Given a set X , a σ -algebra $\mathcal{F} \subseteq 2^X$ is a non-empty collection of subsets closed under complementation and countable unions (hence also countable intersections).

Definition B.2 (Measurable Space). A *measurable space* is a pair $(X, \mathcal{F})$ where $\mathcal{F}$ is a σ -algebra on X (Definition B.1).

Definition B.3 (Measurable Map). If $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ are measurable spaces, a map $f : X \rightarrow Y$ is *measurable* if $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{G}$.

Definition B.4 (Measure & Probability Measure). Let $(X, \mathcal{F})$ be a measurable space. A *measure* is a countably additive map $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$. It is a *probability measure* if $\mu(X) = 1$. We write PX for the set of all probability measures on $(X, \mathcal{F})$.

Definition B.5 (Push-Forward Measure). Given a measurable $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $\mu \in PX$, the *push-forward* measure $f_*\mu \in PY$ is defined by $(f_*\mu)(B) = \mu(f^{-1}(B))$ for $B \in \mathcal{G}$.

Definition B.6 (Pull-Back Measure). Given a measurable $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $\mu \in PY$, the *pull-back* measure $f^*\mu \in PX$ is defined by $f^*\mu(A) := \mu(f(A))$ for $A \in \mathcal{F}$.

B.2 Category Theory

Definition B.7 (Category). A *category* $\mathcal{C}$ consists of a class of objects, hom-sets $\text{Hom}_{\mathcal{C}}(A, B)$ for each ordered pair (A, B) , identity morphisms id_A , and associative composition, satisfying the usual unit laws.

Definition B.8 (Functor). Let $\mathcal{C}, \mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns to each object A of $\mathcal{C}$ an object FA of $\mathcal{D}$ and to each morphism $f : A \rightarrow B$ a morphism $Ff : FA \rightarrow FB$, preserving identities and composition.

Definition B.9 (Finite products in a category). Let $\mathcal{C}$ be a category and $\{X_i\}_{i=1}^n$ a finite family of objects in $\mathcal{C}$ ($n \geq 0$).

- A *product* of the family is a pair

$$\left(X, \{\pi_i\}_{i=1}^n\right), \quad \pi_i : X \longrightarrow X_i \text{ in } \mathcal{C},$$

such that for every object Y and every collection of morphisms $\{f_i : Y \rightarrow X_i\}_{i=1}^n$ there exists a *unique* morphism

$$f : Y \longrightarrow X \quad \text{with} \quad \pi_i \circ f = f_i \quad (i = 1, \dots, n).$$

- When it exists, the product is unique up to a unique isomorphism; we write $X = \prod_{i=1}^n X_i$ and call the π_i the *projection morphisms*.
- For $n = 0$ the empty product is any *terminal object* $1 \in \mathcal{C}$ (unique up to unique isomorphism).

$$\begin{array}{ccccc} & & Y & & \\ & \swarrow f_1 & \downarrow f & \searrow f_2 & \\ X_1 & \xleftarrow{\pi_1} & X_1 \times X_2 & \xrightarrow{\pi_2} & X_2 \end{array}$$

The dashed arrow f is the *unique* morphism rendering all triangles commutative: $\pi_i \circ f = f_i$ for $i = 1, \dots, n$. This diagram visualises the universal property that characterises the product object $X = \prod_{i=1}^n X_i$.

Definition B.10 (Natural Transformation). Given functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a *natural transformation* $\eta : F \Rightarrow G$ assigns to every object A a morphism $\eta_A : FA \rightarrow GA$ such that $Gf \circ \eta_A = \eta_B \circ Ff$ for all $f : A \rightarrow B$ in $\mathcal{C}$.

Definition B.11 (Monad). A *monad* (T, η, μ) on a category $\mathcal{C}$ comprises an endofunctor $T : \mathcal{C} \rightarrow \mathcal{C}$, a unit natural transformation $\eta : \text{id}_{\mathcal{C}} \Rightarrow T$ and a multiplication $\mu : T^2 \Rightarrow T$ satisfying $\mu \circ T\eta = \mu \circ \eta T = \text{id}_T$ and $\mu \circ T\mu = \mu \circ \mu T$.

Definition B.12 (Category **Meas**). Objects are measurable spaces; morphisms are measurable maps (Definition B.3).

Definition B.13 (Markov Kernel). Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces. A *Markov kernel* $\kappa : X \rightsquigarrow Y$ is a map $\kappa : X \times \mathcal{G} \rightarrow [0, 1]$ such that

1. for each $x \in X$, $B \mapsto \kappa(B \mid x)$ is a probability measure on $(Y, \mathcal{G})$;
2. for each $B \in \mathcal{G}$, $x \mapsto \kappa(B \mid x)$ is $\mathcal{F}$ -measurable.

Definition B.14 (Category **Stoch**). Objects are measurable spaces. Morphisms are Markov kernels (Definition B.13). Composition is *Chapman–Kolmogorov* integration:

$$(\kappa * \lambda)(C \mid x) = \int_Y \kappa(C \mid y) \lambda(dy \mid x).$$

Definition B.15 (Giry Monad). Define $G : \mathbf{Meas} \rightarrow \mathbf{Meas}$ by $G(X, \mathcal{F}) = (PX, \mathcal{PF})$, where $\mathcal{PF}$ is the weak σ -algebra generated by $\mu \mapsto \mu(B)$ ($B \in \mathcal{F}$). For $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ set $Gf = f_*$ (Definition B.5). The *unit* $\eta_X : X \rightarrow GX$ is $x \mapsto \delta_x$ and the *multiplication* $\mu_X : GGX \rightarrow GX$ is $\mu_X(\Phi)(B) = \int_{PX} \nu(B) d\Phi(\nu)$. With these, (G, η, μ) is a monad on **Meas** (Definition B.11); its Kleisli category is **Stoch** (Definition B.14).

Definition B.16 (Stochastic Dynamical System). A *stochastic dynamical system* is a G -coalgebra: a measurable map $c : X \rightarrow GX$ (Definition B.15). Iterating c via Kleisli composition yields a family of kernels $c^{(n)} : X \rightsquigarrow X$; see Definition B.17.

B.3 Markov Kernels & Stochastic Processes

Definition B.17 (Transition Kernel Family). A family $\{\kappa_t\}_{t \geq 0}$ of kernels $\kappa_t : X \rightsquigarrow X$ is a *transition kernel family* if

$$\kappa_{t+s} = \kappa_t * \kappa_s, \quad \kappa_0(x, \cdot) = \delta_x.$$

Definition B.18 (Discrete-Time Markov Chain). A *discrete-time Markov chain* on state space X is a pair (κ, μ_0) where $\kappa : X \rightsquigarrow X$ is a kernel and $\mu_0 \in PX$ an initial distribution. The law of X_{n+1} given $X_n = x$ is $\kappa(\cdot \mid x)$.

B.4 Semigroup & Generator Theory

Definition B.19 (Bounded Linear Operators). For a Banach space B , the set $\mathcal{L}(B)$ consists of all linear maps $A : B \rightarrow B$ with finite operator norm $\|A\|_{\mathcal{L}(B)} := \sup_{f \neq 0} \frac{\|Af\|_B}{\|f\|_B}$

Equipped with this norm, is itself a Banach space.

Definition B.20 (Operator Semigroup). Let B be a Banach space. A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is called an *operator semigroup* (on B) if

$$P_0 = \text{id}_B, \quad P_{t+s} = P_t P_s \quad \text{for all } s, t \geq 0.$$

Definition B.21 (Uniformly Continuous Semigroup). A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is a *uniformly continuous semigroup* if

$$\lim_{t \downarrow 0} \|P_t - \text{id}\|_{\mathcal{L}(B)} = 0, \quad P_0 = \text{id}, \quad \text{for all } t \geq 0$$

Equivalently, $P_t = e^{tA}$ for a *bounded* operator $A \in \mathcal{L}(B)$ called the (bounded) generator, and the map $t \mapsto P_t$ is continuous in operator norm.

Definition B.22 (Strongly Continuous Semigroup). A family $\{P_t\}_{t \geq 0} \subset \mathcal{L}(B)$ is a *strongly continuous semigroup* (C_0 -semigroup) if

$$\lim_{t \downarrow 0} \|P_t f - f\|_B = 0, \quad \text{for all } f \in B, t \geq 0$$

Definition B.23 (Markov Semigroup). A C_0 -semigroup $\{P_t\}_{t \geq 0}$ on $L^p(X, \lambda)$ is a *Markov semigroup* if each P_t is positive and $P_t 1 = 1$.

Definition B.24 (Infinitesimal Generator). For a C_0 -semigroup $\{P_t\}_{t \geq 0}$ the *generator* is the (possibly unbounded) operator

$$Lf = \lim_{t \downarrow 0} \frac{P_t f - f}{t},$$

with domain $D(L) = \{f : \text{limit exists in } \|\cdot\|\}$.

#	Block / Item	Purpose (what it secures)
1.1	Definition C^k flow on a compact manifold	Fixes baseline regularity (first/second jets exist)
1.2	Definition category Dyn	Formal language for smooth dynamical systems
1.3	Remark equiv. $\Leftrightarrow$ dynamics-preserving	Bridges intuition with formalism
1.4	Example torus $\rightarrow$ circle projection	Concrete morphism in Dyn
1.5	Remark suspension functor Σ	Embeds discrete maps into Dyn
2.1	Lemma finite products in Dyn	Needed for coupled systems / frame bundles
2.2	Lemma coproducts (disjoint unions)	Useful for multi-regime models
2.3	Lemma regular equalisers	Selecting invariant sub-manifolds
2.4	Remark non-existence of arbitrary small limits/colimits	Honesty checkpoint on completeness
3.1	Definition category Vect	Linear target world
3.2	Definition embedding functor $F : \mathbf{Dyn} \rightarrow \mathbf{Vect}$	Moves flows to coordinate data
3.3	Lemma F preserves identities & composition	Ensures commutative squares
3.4	Diagram square for any Ψ	Visual glue between categories
4.1	Definition bandwidth θ , weight matrix W_x	Specifies locality
4.2	Definition LGWLS linear patch B_x	First-order local model
4.3	Definition weighted residuals R_x , covariance C_x	Captures curvature
4.4	Lemma smoothness of (B_x, C_x) (spectral gap)	Pre-req for Rose kernel
4.5	Definition residual eigen-basis Q_x	Second-order directions (frame field)
5.1	Definition Rose kernel ρ_θ	Stochastic local transition
5.2	Definition Rose operator $\mathcal{R}_\theta$	Main spectral object
5.3	Lemma $\mathcal{R}_\theta$ is Markov	Positivity & mass preservation
5.4	Theorem $\mathcal{R}_\theta$ is Hilbert–Schmidt	Compactness $\Rightarrow$ discrete spectrum
5.5	Theorem perturbation vs. Peron–Frobenius	Links Rose to classical operator
5.6	Corollary invariant density h_θ	Statistical meaning
5.7	Theorem continuous-time generator limit	Connects to Fokker–Planck form
6.1	Lemma $F(\Psi) \mathcal{R} = \mathcal{R} F(\Psi)$	Analyses commute with morphology
6.2	Nat. transf. $K \Rightarrow \mathcal{R}_\theta$	Minimal noise extension of Koopman
6.3	Remark pipeline $N \circ F$ mirrors software modules	Practical coding insight
7.1	Algorithmic block fit / basis-update / projection	Implementation of LGWLS + Rose
7.2	Result complexity & caching strategy	Performance guarantees
7.3	Diagnostic empirical smoothness of Q_x	Validates frame field numerically

Table 1: Logical roadmap: definitions, lemmas, and proofs needed to connect geometric, categorical, operator-theoretic, and numerical perspectives.